



# PROJECT-BASED LEARNING

## PHASE-I EVALUATION

### FOOD RESCUE

Smart Food Rescue and Redistribution System

Team ID: T381

**Department of Computer Science & Engineering**

Graphic Era (Deemed to be University), Dehradun

Academic Session 2026-27

# Team & Mentor Details

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## Team Information

**Team ID:** T381  
**section:** ML2 , ML3 , DS1 , ML4  
**Semester:** 3rd  
**Domain:** SOCIAL IMPACT

## Team Members

1. ADITYA DUTTA (Lead) – 2510373905
2. UTSAV BAJORIA – 2510370581
3. VISHWAJEET SINGH – 2510380298
4. RAGHAV PANDEY – 2510380113

## Mentor

Mr. CHIRAG TYAGI

**Interactions completed: 3**

# Problem Statement & Motivation

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## Problem Statement

Restaurants, hostels, households, and events often generate surplus food that is still safe to consume, while NGOs and people in need may face shortages. Without proper coordination, this usable food goes to waste. FoodRescue connects donors with NGOs, recording food quantity, location, type, and expiry, and prioritizing by expiry, quantity, and distance.

## Motivation

- ▶ Restaurants, hostels, households, and events generate surplus food that's still safe to eat
- ▶ Without proper coordination, this usable food simply goes to waste
- ▶ NGOs and people in need often face shortages at the same time

## Target Users / Stakeholders

- ▶ Food donors: restaurants, hostels, households, event organizers
- ▶ NGOs and people in need (recipients)
- ▶ Volunteers responsible for delivery

# Objectives

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## Project Objectives

1. understand the requirements and design the main entities and classes.
2. manage donors, food donations, requirements, and volunteers.
3. select donations by priority and match donors with NGOs.
4. design graph of location and find the best delivery route using Dijkstra's algorithm.
5. Add delivery tracking, test system and check whether food getting delivered to people in need or not.

**Key Objective:** A working C++ (OOP + DSA) system connecting food donors with NGOs for redistribution.

# Proposed Solution

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## Proposed Approach

1. Donor registers a FoodDonation (type, qty, expiry, location)
2. Priority queue ranks urgency; queue handles NGO requests
3. Hash table: fast donor/NGO/volunteer lookup
4. Graph + Dijkstra's algorithm: best delivery route
5. Delivery tracked from registration to drop-off

## Key Features

- ▶ Web front-end (HTML/CSS/JS) for all 3 roles
- ▶ C++ backend: Donor, FoodDonation, NGO, Volunteer, Requirement, Delivery classes
- ▶ Priority-based donation selection & matching
- ▶ Dijkstra-based route planning
- ▶ Delivery tracking + file-based storage

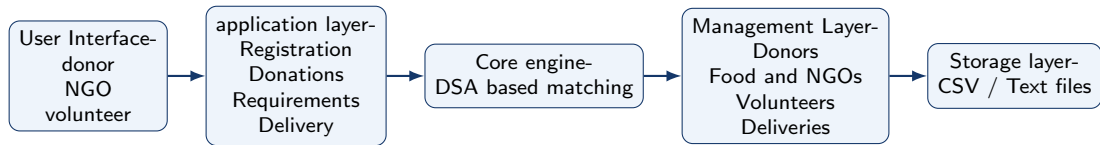
# Integration of PBL Course Concepts

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| Course               | Concepts Applied                                                                                                                                                                                  | Project Module                          |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| Data Structures in C | Priority queue for urgent-donations; using queue for NGO requests; we are using hash table for fast donor/NGO/volunteer lookup; Graph and Dijkstra's algorithm together for better delivery route | DSA-Based Matching & Optimization Layer |
| OOPs with C++        | Classes: Donor, FoodDonation, NGO, Volunteer, Requirement, Delivery; Inheritance & Polymorphism for role-specific behavior; STL containers; File handling for persistence                         | Application / Data Management Layer     |

**Requirement:** Demonstrate meaningful integration of both subjects in **one** project.

# System Architecture / Workflow



## Major Components

- ▶ **UI Layer:** Donors, NGO's, volunteers interact with online system.
- ▶ **Logic Engine:** Constraint checker & allocation logic.
- ▶ **Data Management:** keeps track of all donor, NGO, volunteer delivery records

## Data / Control Flow

- ▶ Donor lists a food donation (type, qty, expiry, location)
- ▶ Engine ranks by urgency, matches to a suitable NGO, routes via Dijkstra's algorithm
- ▶ Volunteer assigned; delivery tracked to storage

# Technology Stack & Methodology

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## Technology Stack

- ▶ Backend: C++ (STL, OOP, DSA)
- ▶ Frontend: HTML, CSS, JavaScript
- ▶ Storage: File-based (CSV / Text records)
- ▶ Tools / Framework: Standard Template Library (STL)
- ▶ IDE: VS Code / Code::Blocks
- ▶ Version Control: Git / GitHub

## Development Methodology

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Evaluation



# Initial Progress & Team Contribution

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## Progress Achieved

- ▶ Literature/background study on food-rescue platforms completed
- ▶ Requirements and user roles finalized
- ▶ System architecture and class diagram prepared

## Individual Contribution

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| Member           | Module                       |
|------------------|------------------------------|
| Aditya Dutta     | Core-DSA, Integration & UI   |
| Utsav Bajoria    | Volunteer & Delivery Module  |
| Vishwajeet Singh | NGO & Requirement Module     |
| Raghav Pandey    | Donor & Food Donation Module |

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# Roadmap, Expected Outcomes & References

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## Roadmap

1. Requirements & core entity/class design
2. Donor, volunteer management
3. Priority-based selection
4. best delivery route
5. Delivery tracking & testing

## Expected Outcomes

- ▶ automated priority based matching
- ▶ To cut food wastage and get surplus meals to people who need them
- ▶ GPS backed mobile platform serving multiple cities and NGO's

## References

1. Cormen, T.H., Leiserson, C.E., Rivest, R.L., and Stein, C. – *Introduction to Algorithms*
2. Balagurusamy, E. – *OOPs with C++*
3. C++ Standard Template Library (STL) documentation
4. lecture notes on Data Structures and Object-Oriented Programming